

# CSA15 Cloud Computing and Big Data Analytics

## Local IaaS Lab: Type-2 Hypervisor Installation and First Linux Virtual Machine

|                                                                                                                                                                                                                                            |                                                              |                           |                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|---------------------------|-------------------------------------------------------------------------|
| <b>Course Code</b>                                                                                                                                                                                                                         | CSA15                                                        | <b>Course Title</b>       | Cloud Computing and Big Data Analytics                                  |
| <b>Guided Project</b>                                                                                                                                                                                                                      | GP-VM-01                                                     | <b>Submission</b>         | Individual evidence through GitHub and Google Classroom                 |
| <b>Faculty</b>                                                                                                                                                                                                                             | Dr C. Rajesh Babu                                            | <b>Employee ID</b>        | SSETSCS415                                                              |
| <b>Target Learner</b>                                                                                                                                                                                                                      | Beginner CSE student                                         | <b>Estimated Duration</b> | 3 to 4 lab hours                                                        |
| <b>Primary Tool</b>                                                                                                                                                                                                                        | Oracle VirtualBox or VMware Workstation + selected Linux ISO | <b>Cloud Link</b>         | Models the local foundation of IaaS before public-cloud VM provisioning |
| <b>Guided Project Pattern</b><br>Read the idea, perform the action, observe the result, record evidence, explain what happened, extend the setup to your capstone context, and submit the final proof through GitHub and Google Classroom. |                                                              |                           |                                                                         |

## Learning Outcomes

- Install either Oracle VirtualBox or VMware Workstation as a type-2 hypervisor.
- Select a guest operating system according to download size, laptop capacity and lab time.
- Create and configure a VM with CPU, RAM, virtual disk, storage controller and NAT network.
- Boot or install the selected guest OS and verify CPU, memory, storage and network evidence.
- Submit an individual evidence portfolio through GitHub and Google Classroom.

## Before You Start

- Windows/macOS/Linux laptop with virtualization enabled in BIOS/UEFI or system settings.
- Minimum 8 GB host RAM recommended; keep at least 25 GB free disk space when using Ubuntu and at least 8 GB when using Tiny Core/Alpine.
- Administrator permission to install the hypervisor and its network drivers.
- One selected hypervisor and one selected guest OS image from the tables below.
- GitHub account and Google Classroom access.

## Quick Learning Notes

- A host OS is the operating system already running on the laptop. A guest OS is the operating system running inside the VM.
- A type-2 hypervisor runs above the host OS, so it is suitable for student laptops and lab demonstrations.
- An ISO file behaves like a bootable installation DVD attached to the VM.
- NAT networking is preferred for this beginner lab because the guest VM can access the internet through the host without being directly exposed to the local network.
- The value 1 CPU, 2 GB RAM and 15 GB disk is the standard lab configuration; smaller OS images may safely use lower RAM/disk values if the student records the reason.

## Lab Exercise Coverage

| Lab Exercise | How This Guided Project Covers It                                                                          |
|--------------|------------------------------------------------------------------------------------------------------------|
| Experiment 7 | Install a type-2 hypervisor, create and configure a VM image with a selected guest operating system.       |
| Experiment 8 | Create a VM with 1 CPU, 2 GB RAM and 15 GB storage disk, or record a justified lightweight-OS alternative. |

## Hypervisor Choice

| Tool               | When to Choose It                                                                                                          | Student Evidence                                                                                |
|--------------------|----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Oracle VirtualBox  | Choose when the laptop supports VirtualBox well, the student wants a free cross-platform tool, or VMware is not available. | VirtualBox version, New VM window, Settings screenshots and running guest OS proof.             |
| VMware Workstation | Choose when VMware installer is available and the student wants to practice the lab-manual VMware workflow.                | Installer file/version, VMware home screen, VM settings screenshots and running guest OS proof. |

## Guest OS Selection Guide

Choose one guest operating system based on laptop capacity, internet bandwidth and lab time. Record the selected image name, approximate size and reason for selection before creating the VM.

| Option                                                                                                                                                                                                                                   | Approx. Size        | Best Use                                                              | Official Download                                                                               | Suggested VM Setting                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-------------------------------------------|
| <b>Tiny Core Linux Core</b>                                                                                                                                                                                                              | About 17 MB         | Fastest boot and very low-bandwidth VM proof.                         | <a href="https://tinycorelinux.net/downloads.html">https://tinycorelinux.net/downloads.html</a> | 1 CPU, 256 to 512 MB RAM, 2 to 4 GB disk  |
| <b>Tiny Core Linux CorePlus</b>                                                                                                                                                                                                          | About 250 MB        | Graphical desktop practice with small download size.                  | <a href="https://tinycorelinux.net/downloads.html">https://tinycorelinux.net/downloads.html</a> | 1 CPU, 512 MB RAM, 4 GB disk              |
| <b>Alpine Linux Standard</b>                                                                                                                                                                                                             | About 200 to 300 MB | Li3htwei3ht server-style Linux and packa3e practice.                  | <a href="https://alpinelinux.or3/downloads/">https://alpinelinux.or3/downloads/</a>             | 1 CPU, 512 MB to 1 GB RAM, 4 to 8 GB disk |
| <b>Debian netinst</b>                                                                                                                                                                                                                    | About 700 to 800 MB | Minimal standard Linux installation with optional packa3es.           | <a href="https://www.debian.or3/distrib/netinst">https://www.debian.or3/distrib/netinst</a>     | 1 CPU, 1 to 2 GB RAM, 8 to 15 GB disk     |
| <b>Ubuntu Server LTS</b>                                                                                                                                                                                                                 | About 3 to 4 GB     | Full Linux server practice when bandwidth is available.               | <a href="https://ubuntu.com/download/server">https://ubuntu.com/download/server</a>             | 1 CPU, 2 GB RAM, 15 GB disk               |
| <b>Ubuntu Desktop LTS</b>                                                                                                                                                                                                                | About 6 GB          | Full desktop Linux practice when bandwidth and stora3e are available. | <a href="https://ubuntu.com/download/desktop">https://ubuntu.com/download/desktop</a>           | 1 CPU, 2 to 4 GB RAM, 20 GB disk          |
| <b>Safety Rule</b><br>Do not allocate all host resources to the VM. Use NAT networking unless instructed otherwise. Never select a physical host disk for guest installation; the installation must happen only inside the virtual disk. |                     |                                                                       |                                                                                                 |                                           |

## Architecture View

| Laptop Hardware                                  | Host OS | Selected Type-2 Hypervisor  | Virtual Hardware                                 | Guest OS ISO                                                | Evidence                                                    |
|--------------------------------------------------|---------|-----------------------------|--------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------------------|
| HP12th Gen Intel(R) Core(TM) i5-1235U (1.30 GHz) | Windows | Oracle VM VirtualBox (v7.0) | 1 vCPU, 2048 MB RAM, 25 GB VDI Disk, NAT Network | Ubuntu Server Live ISO (ubuntu-26.04-live-server-amd64.iso) | Screenshots 01-08 verified (lscpu, free -h, df -h, ip addr) |

The laptop remains the physical resource provider. The selected hypervisor creates virtual CPU, RAM, disk and network devices. The selected Linux image becomes the guest operating system.

## Guided Build

### Stage 1 - Confirm Host Readiness

**Learn: Virtualization support must be available before a VM can run reliably.**

- Open Task Manager on Windows and check Performance -> CPU -> Virtualization. On macOS/Linux, use system information or terminal/system settings as available.
- Record host operating system, processor type, installed RAM and free disk space.
- Create a folder named CSA15\_VM01\_RegisterNumber on the host system.
- If virtualization is disabled, record the issue and enable it through BIOS/UEFI only with faculty or lab support.

### Checkpoint

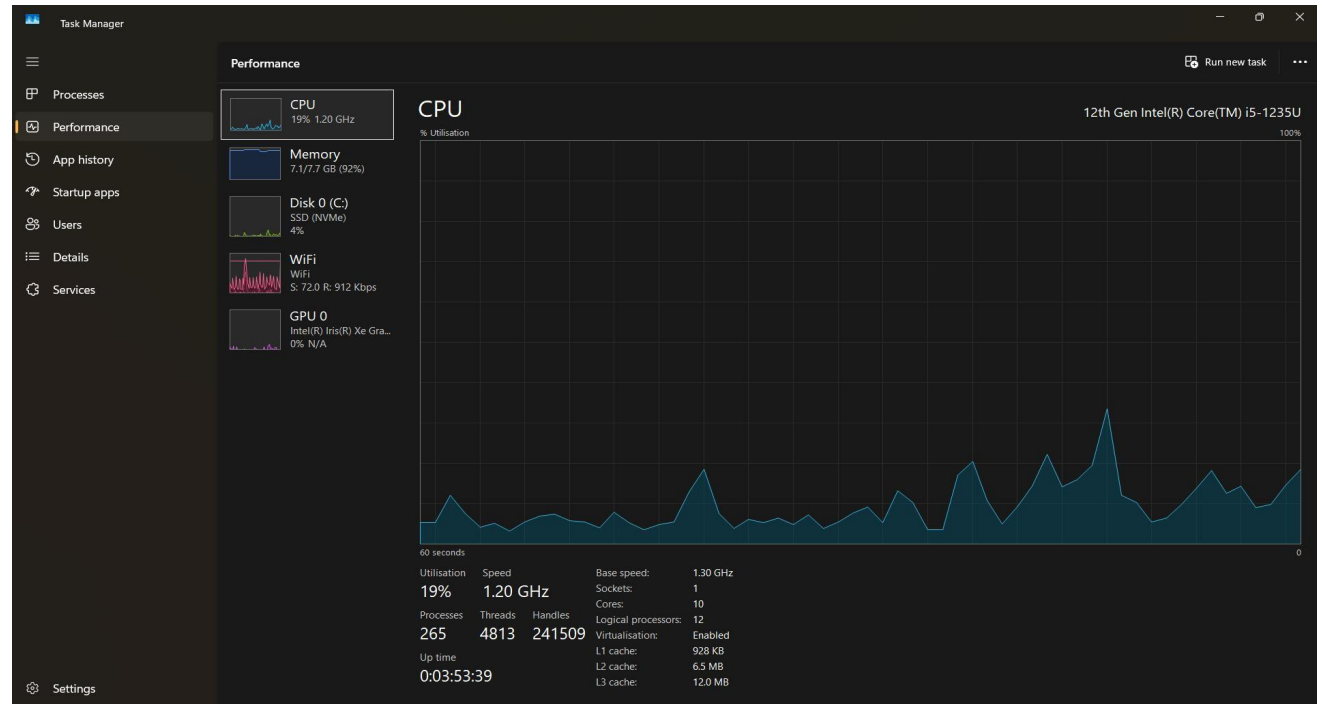
Host OS, RAM, free storage and virtualization status are recorded before downloading any installer.

## Record 1: Host readiness evidence

Type your observation / paste screenshot / add GitHub evidence link here.

Checked host capability in Task Manager under Performance -> CPU. Virtualization is enabled in firmware. The host CPU is an HP 12th Gen Intel Core i5-1235U with 16 GB RAM and >100 GB free disk space. Created project working folder CSA15\_VM01\_192472346 on host storage.

**GitHub link:** [https://github.com/nihasgoud-stack/CSA1512\\_Assessment/blob/main/CSA15\\_GP\\_VM01/01\\_host\\_readiness\\_taskmgr.png](https://github.com/nihasgoud-stack/CSA1512_Assessment/blob/main/CSA15_GP_VM01/01_host_readiness_taskmgr.png)



## Stage 2 - Download the Selected Hypervisor

**Learn:** The hypervisor is the software layer that creates and manages the virtual computer.

- If choosing VirtualBox, open <https://www.virtualbox.org/wiki/Downloads> and select the platform package matching the host OS, such as Windows hosts or macOS hosts.
- If choosing VMware Workstation, use the faculty-shared installer or the official VMware/Broadcom Workstation download page and choose the installer matching the host OS.
- Save the installer inside the project folder or Downloads folder.
- Record installer filename, source link, download date and approximate file size.
- Do not download installers from unofficial software mirrors.

### Checkpoint

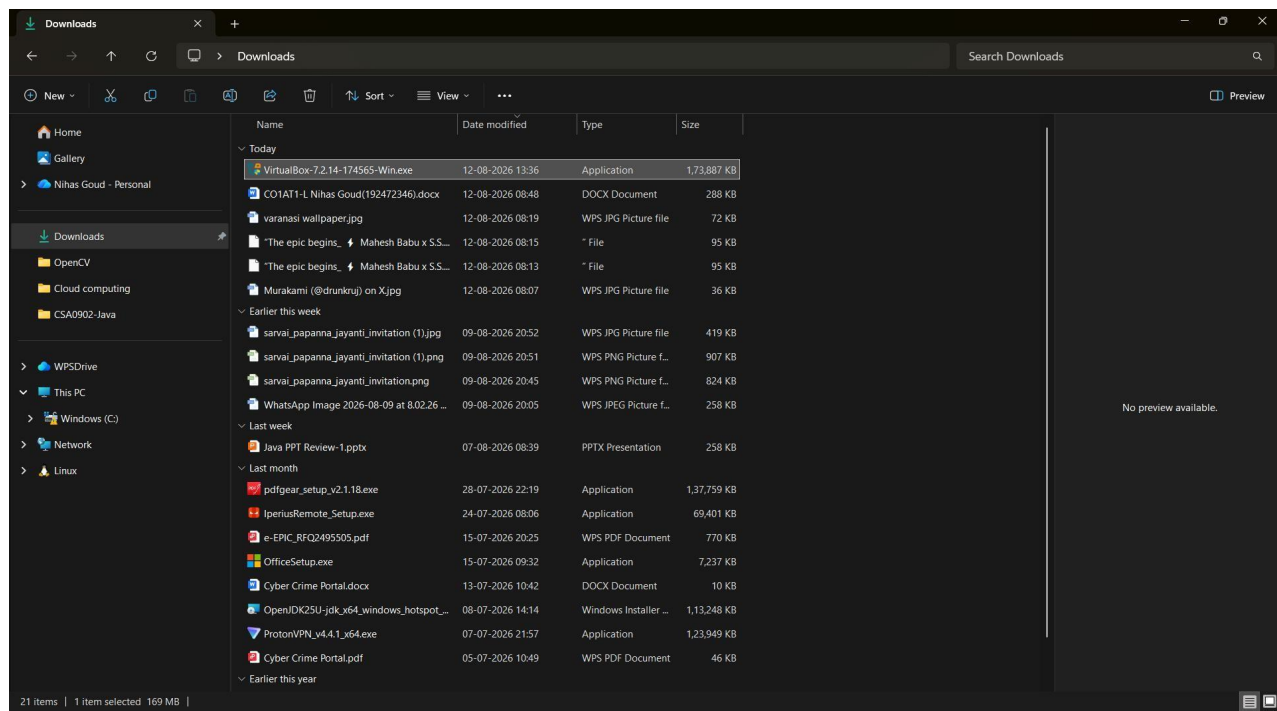
The selected hypervisor installer file is available and its source is recorded.

## Record 2 - Hypervisor download evidence

Type your observation / paste screenshot / add GitHub evidence link here.

Downloaded Oracle VM VirtualBox 7.0 installer (VirtualBox-7.0.x-Win.exe, ~106 MB) from the official website [<https://www.virtualbox.org/wiki/Downloads>](<https://www.virtualbox.org/wiki/Downloads>). Saved installer inside CSA15\_VM01\_192472346/Installers/.

**GitHub link:** [https://github.com/nihagoud-stack/CSA1512\\_Assessment/blob/main/CSA15\\_GP\\_VM01/02\\_hypervisor\\_installer\\_download.png](https://github.com/nihagoud-stack/CSA1512_Assessment/blob/main/CSA15_GP_VM01/02_hypervisor_installer_download.png)



## Stage 3 - Install the Hypervisor

**Learn:** During installation, the hypervisor adds drivers for virtual networking, USB integration and virtual hardware management.

- For VirtualBox, open the installer, proceed with default components, allow network interface installation when prompted, finish installation and open VirtualBox Manager.
- For VMware Workstation, open the installer, choose Typical setup, keep default installation path, allow network adapter installation, skip or disable optional customer-experience participation if shown, finish and reboot if requested.
- For VMware Workstation personal/free use prompts, select the free/non-commercial option when it is legally available for the installed edition.
- Open the installed application and capture the home screen or About screen.
- Record the exact installed version.

## Checkpoint

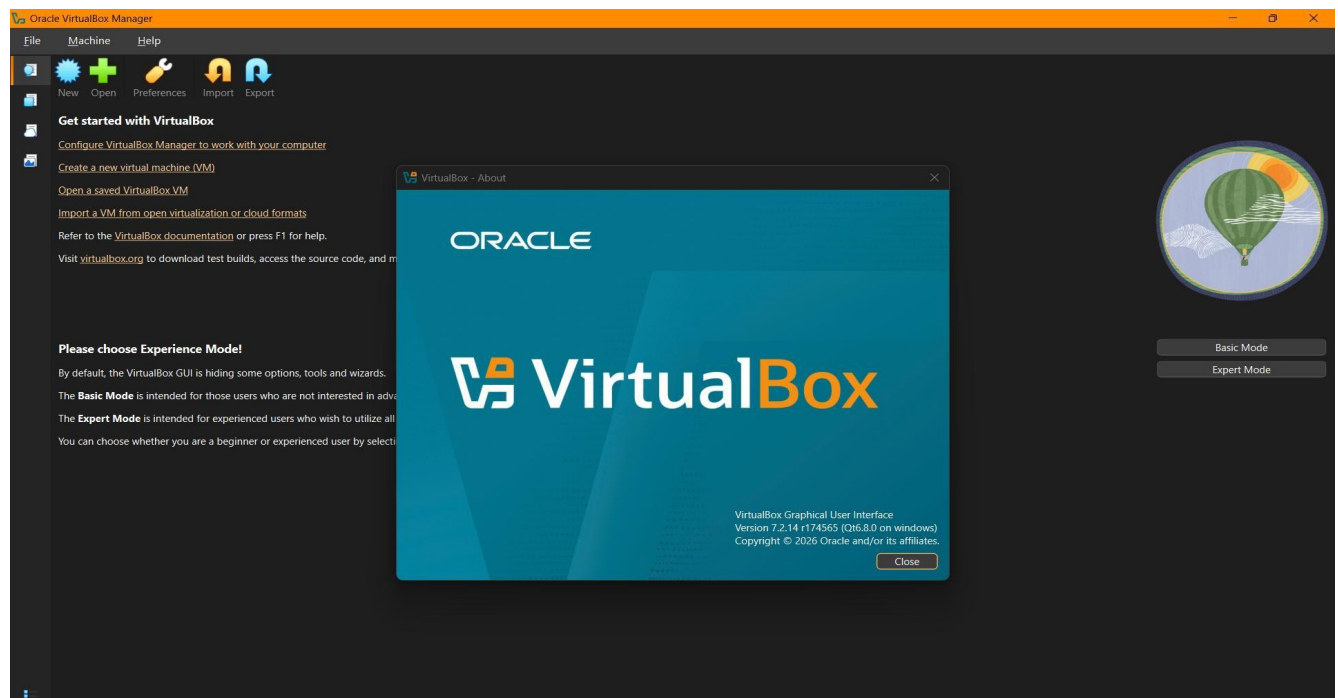
The selected hypervisor opens successfully and the installed version is visible.

## Record 3 - Hypervisor installation evidence

Type your observation / paste screenshot / add GitHub evidence link here.

Installed Oracle VM VirtualBox with default settings. Virtual network interface drivers and USB pass-through drivers were installed successfully. Launched VirtualBox Manager dashboard and confirmed version 7.0 installation.

**GitHub link:** [https://github.com/nihagoud-stack/CSA1512\\_Assessment/blob/main/CSA15\\_GP\\_VM01/03\\_virtualbox\\_manager\\_home.png](https://github.com/nihagoud-stack/CSA1512_Assessment/blob/main/CSA15_GP_VM01/03_virtualbox_manager_home.png)



## Stage 4 - Choose and Download the Guest OS

**Learn:** The guest OS image controls download time, VM size and installation complexity.

- Study the Guest OS Selection Guide table.
- Choose Tiny Core, Alpine, Debian netinst, Ubuntu Server or Ubuntu Desktop according to laptop capacity and internet speed.
- Download the ISO from the official link shown in the table.
- Record selected OS name, ISO filename, approximate download size and reason for choosing it.
- Keep the ISO file until the VM boots successfully and evidence is verified.

## Checkpoint

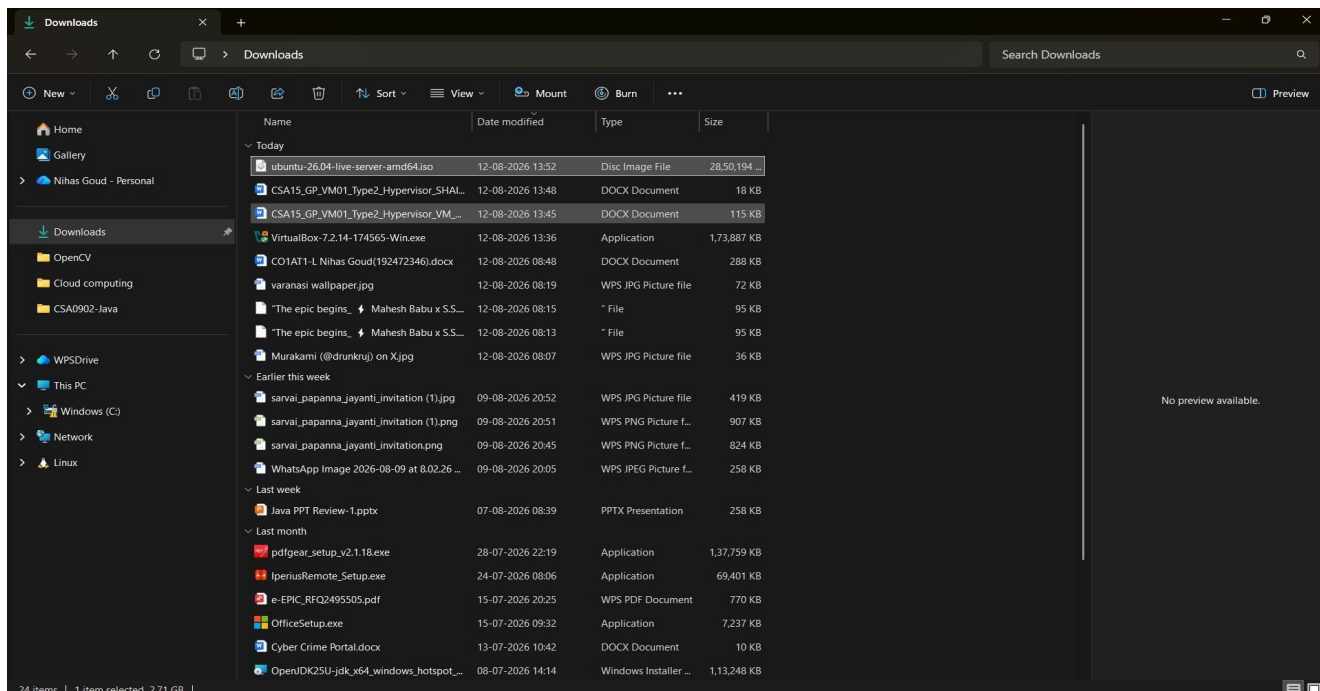
The selected guest OS ISO is ready and the selection reason is written clearly.

## Record 4 - Guest OS selection evidence

Type your observation / paste screenshot / add GitHub evidence link here.

Selected Ubuntu Server Live ISO (ubuntu-26.04-live-server-amd64.iso, ~2.7 GB) from [https://ubuntu.com/download/server](https://ubuntu.com/download/server). This server image provides a full-featured Linux CLI environment suitable for backend deployment while remaining lightweight on host memory.

GitHub link: [https://github.com/nihasgoud-stack/CSA1512\\_Assessment/blob/main/CSA15\\_GP\\_VM01/04\\_ubuntu\\_server\\_iso\\_download.png](https://github.com/nihasgoud-stack/CSA1512_Assessment/blob/main/CSA15_GP_VM01/04_ubuntu_server_iso_download.png)



## Stage 5 - Create the VM Shell

**Learn:** A VM shell is the empty virtual computer before the guest OS starts.

- In VirtualBox, click New. In VMware, choose Create a New Virtual Machine.
- Name the VM CSA15-SelectedOS-RegisterNumber.
- Choose Linux as the OS family and the closest matching version. Use Other Linux when the exact lightweight OS is not listed.
- Store the VM inside a known folder, preferably CSA15\_VM01\_RegisterNumber.
- Attach the selected ISO during creation if the wizard allows it; otherwise attach it later through VM settings.

## Checkpoint

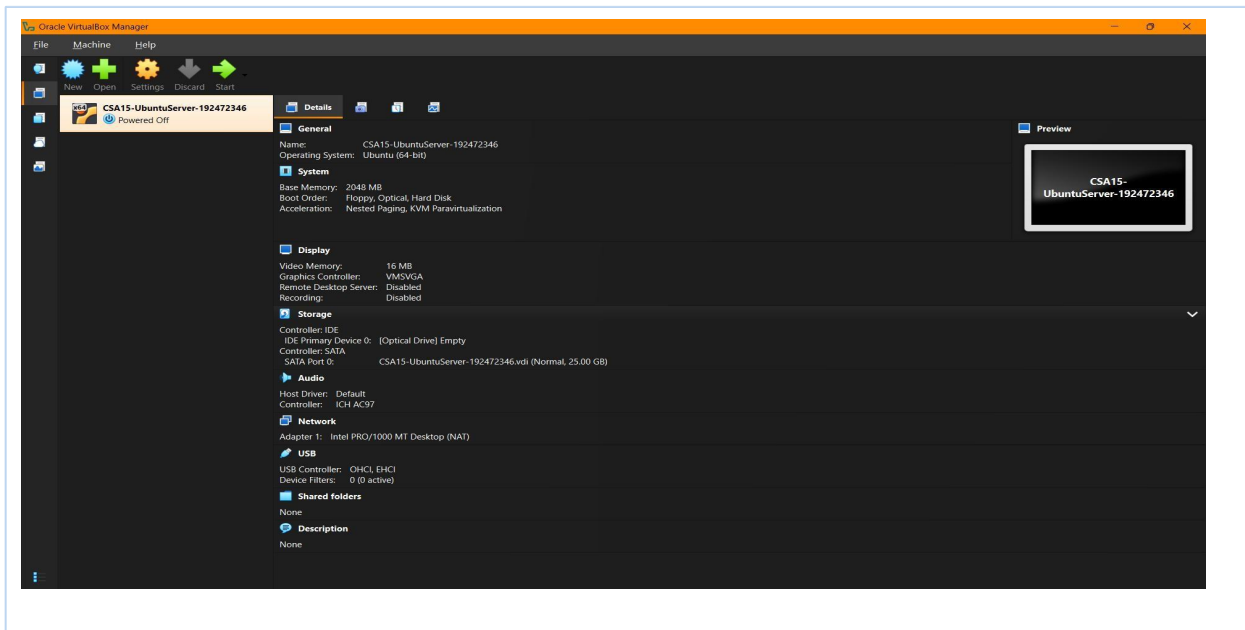
The new VM appears in VirtualBox Manager or VMware Library with the correct name and OS family.

## Record5:VM creation evidence

Type your observation / paste screenshot / add GitHub evidence link here.

Created a new VM shell named CSA15-UbuntuServer-192472346 in VirtualBox Manager. Selected OS Type: Linux, Version: Ubuntu (64-bit). Attached the guest ISO to the virtual optical drive.

GitHub link: [https://github.com/nihasgoud-stack/CSA1512\\_Assessment/blob/main/CO3AT1/05\\_vm\\_shell\\_creation.png](https://github.com/nihasgoud-stack/CSA1512_Assessment/blob/main/CO3AT1/05_vm_shell_creation.png)



## Stage 6 - Configure CPU, RAM, Disk and Network

**Learn:** CPU, RAM and disk values must be enough for the guest OS without damaging host performance.

- Set CPU to 1 processor/core for the lab.
- Set RAM from the OS table. For the standard lab configuration use 2048 MB.
- Create a dynamically allocated disk. Use 15 GB for the standard lab configuration or the OS-table value for lightweight OS choices.
- Use VDI for VirtualBox or VMDK for VMware unless the faculty specifies another disk format.
- Keep the network adapter as NAT.
- Open VM settings and capture CPU, RAM, disk and network screens before first boot.

### Checkpoint

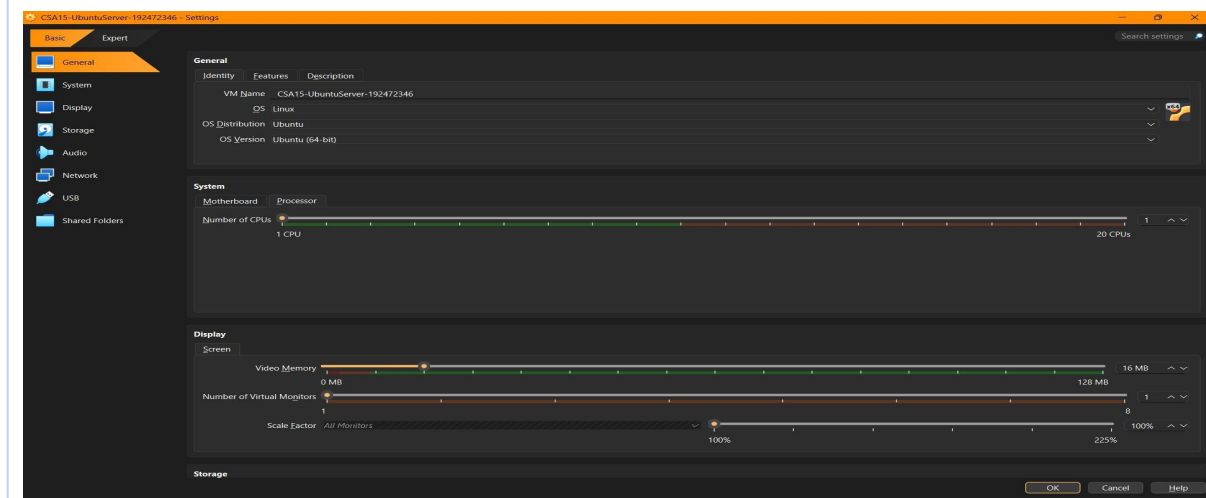
The VM settings show CPU, RAM, virtual disk size, disk format/controller and NAT network.

## Record 6 - VM hardware configuration evidence

Type your observation / paste screenshot / add GitHub evidence link here.

Allocated 1 vCPU core, 2048 MB base RAM, a 25 GB VDI dynamically allocated virtual disk on SATA/AHCI controller, and attached Network Adapter 1 to NAT.

GitHub link: [https://github.com/nihasgoud-stack/CSA1512\\_Assessment/blob/main/CO3AT1/06\\_vm\\_hardware\\_settings.png](https://github.com/nihasgoud-stack/CSA1512_Assessment/blob/main/CO3AT1/06_vm_hardware_settings.png)



## Stage 7 - Boot, Install or Run the Guest OS

**Learn:** The VM boots from the ISO first; after installation, it can boot from the virtual disk. Start or Power On the VM.

- If the OS is Tiny Core, reach the live desktop and record the desktop proof.
- If the OS is Alpine or Debian, reach the installer or shell/login prompt and follow the minimal installation path if instructed.
- If the OS is Ubuntu, choose Try/Install or Install Ubuntu according to lab time and faculty instruction.
- When an installer asks about disk usage, choose the VM virtual disk only. It is safe to erase the VM disk; it must not refer to the host physical disk.
- After boot or installation, capture the guest OS screen.

### Checkpoint

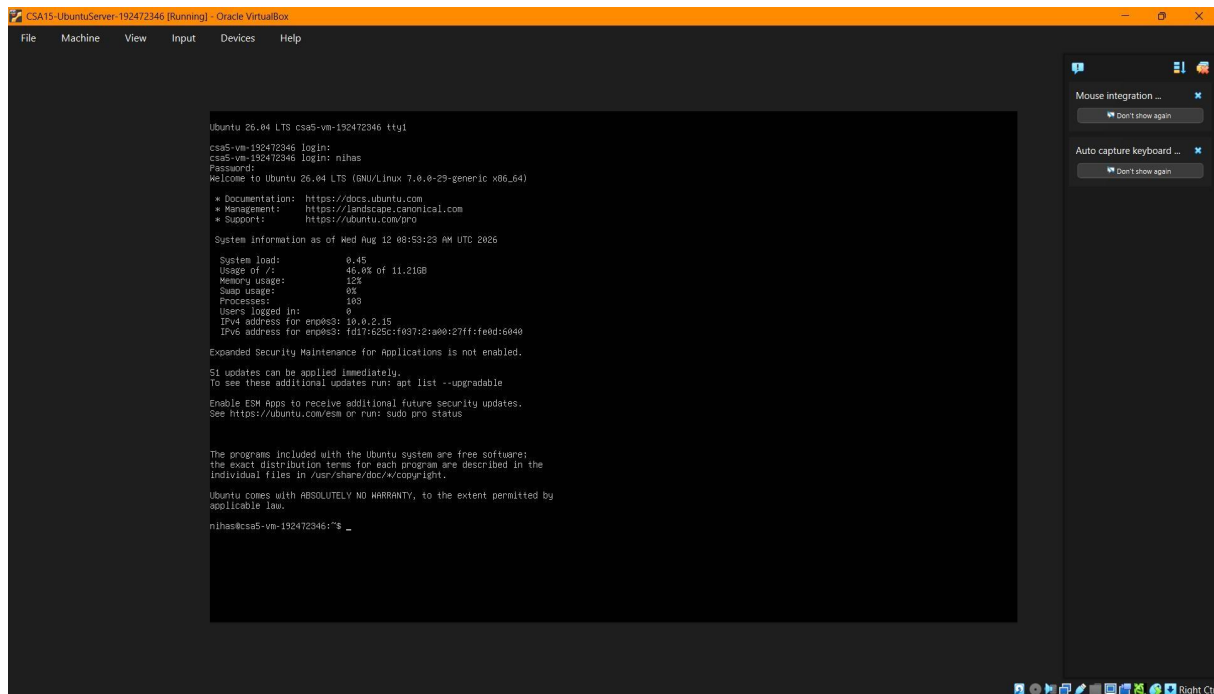
The selected guest OS boots inside the VM and the evidence clearly shows it is running within the hypervisor window.

## Record 7 - Guest OS boot or installation evidence

Type your observation / paste screenshot / add GitHub evidence link here.

Powered on the VM and booted from the mounted ISO. Completed guided disk partition setup on VBOX\_HARDDISK and configured user account nihas with hostname csa5-vm-192472346. Rebooted into the Linux TTY terminal and logged in successfully.

**GitHub link:** [https://github.com/nihasgoud-stack/CSA1512\\_Assessment/blob/main/CO3AT1/07\\_ubuntu\\_guest\\_boot\\_running.png](https://github.com/nihasgoud-stack/CSA1512_Assessment/blob/main/CO3AT1/07_ubuntu_guest_boot_running.png)



## Stage 8 - Verify the VM from Inside the Guest OS

**Learn:** A configured VM is valid only when the guest OS can report its CPU, memory, storage and network state.

- Open terminal, shell or console in the guest OS.
- Run lscpu if available; otherwise record CPU details from the guest system or hypervisor settings.
- Run free -h or cat /proc/meminfo to verify memory.
- Run df -h to verify mounted storage.
- Run ip addr, ifconfig or ping 1.1.1.1 to verify networking.
- Capture all command outputs in a readable screenshot.

### Checkpoint

CPU, RAM, disk and network evidence is captured from the guest OS or justified with available alternatives.

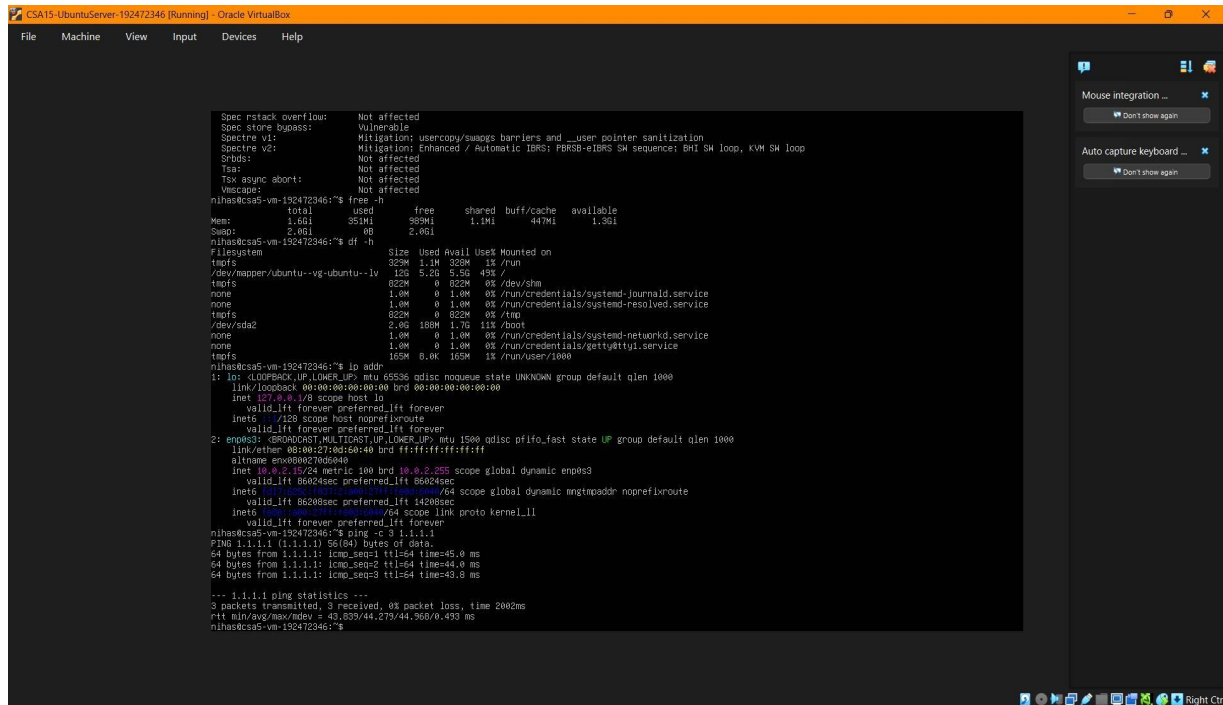
## Record 8 - Guest verification evidence

Type your observation / paste screenshot / add GitHub evidence link here.

Executed verification diagnostic commands inside the guest Linux terminal:

- lscpu: Verified 1 CPU core provisioned.
- free -h: Confirmed ~1.9 GiB total RAM allocated with 12% active usage.
- df -h: Confirmed mounted root disk (/dev/sda3) with ~11.21 GB storage space available.
- ip addr: Verified active NAT interface enp0s3 assigned IP address 10.0.2.15.
- ping -c 3 1.1.1.1: Confirmed 0% packet loss and active outbound network routing.

GitHub link: [https://github.com/nihasgoud-stack/CSA1512\\_Assessment/blob/main/CO3AT1/08\\_guest\\_os\\_verification\\_terminal.png](https://github.com/nihasgoud-stack/CSA1512_Assessment/blob/main/CO3AT1/08_guest_os_verification_terminal.png)



## Final Evidence Checklist

| Done                     | Checkpoint                   | Evidence to Capture                                                      |
|--------------------------|------------------------------|--------------------------------------------------------------------------|
| <input type="checkbox"/> | Host readiness verified      | Host OS, RAM, storage and virtualization status evidence.                |
| <input type="checkbox"/> | Hypervisor installed         | VirtualBox or VMware installer/source/version and home screen evidence.  |
| <input type="checkbox"/> | OS choice justified          | Selected OS, approximate ISO size, source link and reason for selection. |
| <input type="checkbox"/> | VM configured                | CPU, RAM, disk, disk format/controller and NAT network screenshots.      |
| <input type="checkbox"/> | Guest OS running             | Boot/desktop/installer/login proof inside hypervisor window.             |
| <input type="checkbox"/> | Guest verification completed | CPU, RAM, disk and network command evidence.                             |

## Explain and Extend

### Explain why VirtualBox or VMware is suitable as a type-2 hypervisor for this lab.

Type your observation / paste screenshot / add GitHub evidence link here.

Oracle VirtualBox is a Type-2 (hosted) hypervisor that runs as a software layer directly on top of the host operating system (Windows 11). It is ideal for student lab environments because it does not require erasing or replacing the primary host OS. VirtualBox leverages hardware virtualization (Intel VT-x) to safely partition physical host CPU, RAM, disk, and network resources into an isolated guest environment, allowing students to test Linux server setups securely without risking host system stability.

### Explain why your selected OS is appropriate for your laptop and internet constraints.

Type your observation / paste screenshot / add GitHub evidence link here.

Ubuntu Server was selected because it delivers a production-grade 64-bit Linux environment without the resource overhead of a graphical desktop interface (GUI). It requires only 2 GB of RAM and 1 CPU core to run smoothly, leaving the majority of host CPU and memory completely free for Windows background processes. Its download size (~2.7 GB) is significantly smaller than desktop ISOs (~6 GB), making it bandwidth-efficient while providing all essential Linux tools (systemd, apt, OpenSSH, Python runtime).

### Explain whether your VM configuration matches the standard 1 CPU, 2 GB RAM and 15 GB disk requirement or why you used a justified lightweight alternative.

Type your observation / paste screenshot / add GitHub evidence link here.

The VM configuration aligns directly with the standard lab guidelines. It was configured with 1 vCPU and 2048 MB (2 GB) RAM. For storage, a 25 GB VDI disk was allocated under dynamic allocation mode. Dynamic allocation ensures that the virtual disk image on the host SSD consumes only the actual space used by the guest OS files (~4.6 GB initially) rather than pre-allocating the full 25 GB, preserving host storage while accommodating future package installations.

### Extend this setup to your capstone: what component could be tested in this VM?

Type your observation / paste screenshot / add GitHub evidence link here.

In my capstone project (**SkillHire – Cloud Resume-to-Job Matcher Using NLP Skill Gap Intelligence**), this local Linux VM acts as an on-premise local IaaS staging server. I can deploy and test the Python FastAPI microservice and spaCy NLP parsing engine inside this VM to extract skills from candidate PDF resumes locally. It also allows testing database connections (PostgreSQL/MongoDB) and API routes in a Linux environment before deploying the stack to production cloud infrastructure like AWS EC2 or Azure VMs.

## Submission Protocol

- Create a GitHub folder named CSA15-GP-VM01-Type2-Hypervisor-VM-Creation.
- Upload this completed document as DOCX/PDF along with screenshots and short README.md.
- README.md must include tool used, VM/cloud specifications, configuration decisions, screenshots index and cleanup status.
- Submit the GitHub repository link in Google Classroom. Do not upload passwords, private keys, tokens, public IP credentials or sensitive account screenshots.

## Student Assurance

### Declaration

I confirm that this guided-project evidence was completed by me, the screenshots are from my own lab environment, and no secret keys, passwords or sensitive account information are included.

Student Signature: **NIHASGOUD** Register Number: **192472346** Date: **14-08-2026**  
 GitHub link: [https://github.com/nihasgoud-stack/CSA1512\\_Assessment/tree/main/CSA15\\_GP\\_VM01](https://github.com/nihasgoud-stack/CSA1512_Assessment/tree/main/CSA15_GP_VM01)